

Capital Bikeshare Demand Analysis

STAT 24620 / FINM 34700 / STAT 32950 — Group Data Project, Spring 2026

Research Question

Does feature engineering and unsupervised learning-based features help improve supervised prediction of bike-sharing demand?

We investigate whether engineered time-of-day demand profiles and GMM-derived cluster memberships improve prediction accuracy for both hourly rental count (cnt) and high-demand classification (high_demand) compared to raw features alone.

Dataset Description

The Capital Bikeshare Hourly Demand dataset (UCI ML Repository) contains 17,389 hourly observations from the Capital Bikeshare system in Washington D.C. across 2011–2012. Each record includes calendar variables (season, month, hour, weekday, holiday, working day), normalized weather measurements (temperature, humidity, wind speed, weather situation), and rental counts split by casual and registered riders.

Target variables:

- cnt — total hourly rental count (regression)
- high_demand — binary: 1 if cnt $\geq$ 143 (median threshold), else 0 (classification)

Train/test split: 80% train (13,903 rows) / 20% test (3,476 rows), random_state=42.

Preprocessing Decisions

Decision	Rationale
Drop instant, dteday, hour, timestamp	Index/date columns, not informative features
Drop casual, registered as predictors	They sum to cnt — direct data leakage
Drop atemp	0.99 correlation with temp; redundant
Encode high_demand as binary (0/1)	Map Low→0, High→1 for classification
is_weekend flag replacing weekday	Captures weekday vs. weekend signal more cleanly
Sin/cos encoding for hr and mnth	Dense cyclic variables; preserves wrap-around structure
One-hot encoding for season	Only 4 categories; effects are non-sinusoidal
StandardScaler inside pipeline	Required for ElasticNet; prevents scale-dependent penalization
wk_hr_dist / nw_k_hr_dist features	Mean demand by hour $\times$ workingday (note: mild lookahead bias)
GMM cluster one-hot columns	Cluster memberships encoded as binary features (Steps 3+)

Methods

Supervised Methods

ElasticNet (Regression & Classification). A regularized linear model combining L1 (Lasso) and L2 (Ridge) penalties. Alpha and l1_ratio selected via 5-fold cross-validation (ElasticNetCV / LogisticRegressionCV). StandardScaler applied inside a Pipeline. The linear structure makes coefficients directly interpretable but limits the model's ability to capture nonlinear interactions.

Random Forest (Regression & Classification). An ensemble of 100 decision trees (n_estimators=100, random_state=42). No scaling required. Naturally captures nonlinear interactions and feature redundancy. Feature importance derived from mean impurity decrease.

Unsupervised Methods

PCA. Applied to the fully feature-engineered dataset (Step 2.1 features, scaled). Two principal components extracted for visualization. Loading heatmap used to identify dominant variance sources across weather and time features.

Gaussian Mixture Model (GMM). Fitted to hourly demand profiles aggregated by working day. Covariance structures (full, tied, diag, spherical) and component counts (1–29) evaluated by BIC on 5-fold cross-validation of the training split. Optimal cluster count selected at elbow (k=10, full covariance). Cluster memberships one-hot encoded and added as supervised features. A standalone GMM classifier was also evaluated.

Results

Table 1: Regression Results — Predicting cnt

Step	Feature Set	# Feat.	EN MSE	EN R ²	RF MSE	RF R ²
1	Baseline (raw)	12	19,396	0.3875	1,770	0.9441
1.1	+ Cyclic / one-hot encoding	16	15,757	0.5024	2,119	0.9331
2	+ Demand profiles (lookahead)	12	6,885	0.7826	1,970	0.9378
2.1	+ Demand profiles + cyclic	17	6,792	0.7855	1,934	0.9389
3	+ GMM cluster features	17	12,882	0.5932	1,470*	0.9536*
3.1	+ GMM clusters + cyclic	21	11,764	0.6285	1,638	0.9483

* Best RF regression result

Table 2: Classification Results — Predicting high_demand

Step	Feature Set	# Feat.	EN Acc.	EN F1	RF Acc.	RF F1	AUC
1	Baseline (raw)	12	0.782	0.778	0.924	0.923	0.924
1.1	+ Cyclic / one-hot	16	0.841	0.835	0.932	0.931	0.932
2	+ Demand profiles	12	0.905	0.903	0.934	0.933	0.934
2.1	+ Demand prof. + cyclic	17	0.907	0.905	0.932	0.931	0.932
3	+ GMM clusters	17	0.903	0.904	0.948	0.948	0.948
3.1	+ GMM clusters + cyclic	21	0.910	0.910	0.950*	0.949*	0.950*

* Best RF classification result

Table 3: Standalone GMM Classifier Performance

Metric	Value
Precision	0.57
Recall	0.72
F1 Score	0.64

Interpretation of Main Results

Feature engineering strongly benefits ElasticNet

Going from the raw baseline ($R^2 = 0.39$) to demand-profile features ($R^2 = 0.78$) more than doubles explained variance for ElasticNet. This is because ElasticNet cannot learn the nonlinear interaction between hour and working day on its own — the engineered `wk_hr_dist` and `nwk_hr_dist` features encode this interaction explicitly, making it linearly accessible.

Random Forest is robust but also benefits from GMM features

Random Forest achieves high accuracy even on raw features ($R^2 = 0.944$), demonstrating its ability to learn complex interactions natively. However, adding GMM-derived cluster features (Step 3) still improves RF regression to $R^2 = 0.9536$. The best overall classification result (RF Acc. = 0.9497, AUC = 0.9499) was achieved in Step 3.1 with both GMM clusters and cyclic encoding.

PCA reveals time-of-day and weather as dominant variance drivers

The PCA loading heatmap shows that hour-related features (`hr_sin`, `hr_cos`, `wk_hr_dist`) load strongly on one component while weather variables (`temp`, `hum`) load on another. The scatter plot colored by hour of day shows clear circular arcs in PC1–PC2 space, confirming the strong cyclic time-of-day structure.

GMM identifies meaningful demand regimes

The BIC curve identified an elbow at $k=10$ for the full covariance GMM. The 10 clusters map to interpretable demand regimes: high-peak commute hours on working days (cluster 8, mean cnt ≈ 429), moderate afternoon working-day demand, weekend leisure demand, and overnight low-demand hours. As a standalone predictor, the GMM classifier achieves $F1 = 0.64$, well below the supervised RF ($F1 = 0.92+$), confirming that unsupervised structure alone is insufficient and its value is as an augmentation to supervised methods.

Cyclic encoding helps ElasticNet more than RF

Adding sin/cos hour and month encoding (Step 1.1 vs Step 1) improves ElasticNet R^2 from 0.39 to 0.50 while Random Forest R^2 slightly decreases ($0.944 \rightarrow 0.933$). RF can already find correct split thresholds for raw hour values; ElasticNet needs the cyclic structure made explicit.

Comparison of Methods

Dimension	ElasticNet	Random Forest
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Raw baseline (regression)	Moderate ($R^2 \approx 0.39$)	High ($R^2 \approx 0.94$)
Benefit from demand profiles	Very large (+0.39 R^2)	Small (+0.00 R^2)
Benefit from GMM features	Moderate (+0.09 R^2)	Positive (+0.01 R^2)
Cyclic encoding benefit	Large (+0.11 R^2)	Negligible / slight decrease
Interpretability	High — direct coefficients	Moderate — feature importances
Best regression performance	$R^2 = 0.786$ (Step 2.1)	$R^2 = 0.954$ (Step 3)
Best classification AUC	0.910 (Step 3.1)	0.950 (Step 3.1)

The central finding is that feature engineering closes the gap between linear and non-linear models but does not eliminate it. Random Forest consistently outperforms ElasticNet across all steps, but ElasticNet's relative gain from engineered features is far larger. GMM cluster features provide the most reliable incremental improvement across both model families.

Limitations

- **Lookahead bias in demand profile features.** `wk_hr_dist` and `nwk_hr_dist` are computed from the full dataset before the train/test split. The strong ElasticNet improvement in Steps 2 and 2.1 should be interpreted with this caveat.
- **GMM clustering on training data.** Cluster labels assigned to the full dataset use the full-data fitted GMM, which is a mild form of in-sample information.
- **Static demand threshold.** The `high_demand` binary label uses a fixed threshold (143) computed on the full dataset — a form of pre-processing leakage.
- **No temporal cross-validation.** A random split was used. A time-based split (train on 2011, test on 2012) would better reflect real forecasting performance.
- **Untuned Random Forest.** `n_estimators=100` with default depth. Tuning `max_depth`, `min_samples_split`, and `max_features` could further improve RF performance.

Conclusion

Feature engineering and unsupervised learning features both contribute to improved supervised prediction performance, though with different magnitudes by model type. For ElasticNet, demand profile features produce the largest gain by encoding nonlinear time-of-day interactions in a linearly accessible form. For Random Forest, GMM-derived cluster memberships provide the most consistent incremental improvement. The best overall models — RF with GMM clusters and cyclic encoding — achieve $R^2 = 0.9536$ for regression and $AUC = 0.9499$ for classification, compared to $R^2 = 0.9441$ and $AUC = 0.9245$ for the raw baseline. These results support the hypothesis that structured feature engineering and unsupervised demand regime discovery meaningfully improve bike-sharing demand prediction.